ECA02 – DIGITAL CIRCUITS

# CO5 CHARACTERISTICS INTERPRETATION ASSIGNMENT

NMOS, PMOS Transistors and CMOS Inverter

| **Student Details** | |
|---|---|
| Name | Pranesh.R |
| Register Number | 1912260397 |
| Course | ECA02 – Digital Circuits |
| Assignment | CO5 – Characteristics Interpretation |

Prepared as a soft-copy assignment

# Introduction

A MOSFET is a voltage-controlled semiconductor device. The gate voltage controls the current flowing between the drain and source. NMOS and PMOS transistors are used together to build a CMOS inverter. This assignment explains their operating regions, gate capacitance, inverter voltage-transfer characteristics, n-well structure and low-power performance.

**Assumptions:** The supplied assignment gives no measured I–V or capacitance data and no transistor dimensions or process parameters. Therefore, the answer uses standard long-channel MOSFET models and qualitative capacitance behavior. Numerical device-specific values must be taken from the actual fabrication/process data or SPICE model.

# Part (a) – NMOS and PMOS Operating Regions

## 1. Important terminal voltages

For an NMOS, the source is usually near ground and the gate voltage is measured relative to the source: VGS = VG – VS. The drain-to-source voltage is VDS = VD – VS. The threshold voltage is VTN.

For a PMOS, it is easier to use positive magnitudes: VSG = VS – VG and VSD = VS – VD. The threshold magnitude is |VTP|.

## 2. NMOS operating regions

| Region | Condition (NMOS) | Meaning |
|---|---|---|
| Cut-off | VGS < VTN | Channel is not formed; ID is approximately zero. |
| Triode / Linear | VGS > VTN and 0 ≤ VDS < VGS – VTN | Channel is formed; device behaves approximately like a voltage-controlled resis |
| Saturation | VGS > VTN and VDS ≥ VGS – VTN | Channel pinches off near the drain; current is mainly controlled by VGS. |

**Cut-off:** When the gate-to-source voltage is below threshold, the inversion channel is absent. Only a very small leakage current flows.

**Linear region equation:** $ID = \mu n\ Cox\ (W/L)\ [(VGS – VTN)VDS – VDS^2/2]$, for $0 \le VDS < VGS – VTN$.

**Saturation region equation:** $ID = (1/2)\ \mu n\ Cox\ (W/L)(VGS – VTN)^2$, for $VDS \ge VGS – VTN$. This ideal equation ignores channel-length modulation.

Here μn is electron mobility, Cox is gate-oxide capacitance per unit area, W is channel width and L is channel length.

## 3. PMOS operating regions

| Region | Condition (PMOS magnitude form) | Meaning |
|---|---|---|
| Cut-off | VSG < \|VTP\| | No strong channel; current is approximately zero. |
| Triode / Linear | VSG > \|VTP\| and 0 ≤ VSD < VSG – \|VTP\| | A conducting p-channel is formed; device behaves like a controlled resisto |
| Saturation | VSG > \|VTP\| and VSD ≥ VSG – \|VTP\| | Channel pinches off near the drain; current is mainly controlled by VSG. |

**PMOS linear current magnitude:** $|ID| = \mu p\ Cox\ (W/L)\ [(VSG – |VTP|)VSD – VSD^2/2]$.

**PMOS saturation current magnitude:** $|ID| = (1/2)\ \mu p\ Cox\ (W/L)(VSG – |VTP|)^2$.

The conventional PMOS drain current direction is opposite to NMOS under the usual terminal reference direction. The equations above show current magnitude to avoid sign confusion.

### 4. Change from cut-off to saturation

• NMOS starts conducting when VGS rises above VTN.

• At first, if VDS is small, the NMOS operates in the linear region.

• When VDS reaches approximately VGS – VTN, the device enters saturation.

• PMOS starts conducting when VSG rises above |VTP|.

• PMOS first operates in the linear region for small VSD and enters saturation when VSD reaches VSG – |VTP|.

**Interpretation:** Increasing gate overdrive (VGS–VTN for NMOS or VSG–|VTP| for PMOS) increases the channel charge and the drain-current capability.

## Part (b) – Gate Capacitance and MOS Operating Regions

### 1. Why gate capacitance changes

The MOS gate is separated from the semiconductor by a thin insulating oxide. The gate, oxide and semiconductor form a capacitor. When gate voltage changes, the charge under the oxide also changes. The measured gate capacitance therefore depends on the surface condition of the semiconductor.

**Oxide capacitance per unit area:** Cox = εox / tox, where εox is oxide permittivity and tox is oxide thickness. For gate area A, the ideal oxide capacitance is Cox,total = εox A / tox.

### 2. NMOS capacitance regions (p-type body)

| Gate condition | Surface behavior | Capacitance interpretation |
|---|---|---|
| Accumulation (negative VG) | Holes gather near the oxide interface | Measured capacitance is close to Cox. |
| Depletion (VG increased toward positive) | Holes are pushed away; a depletion region forms | Depletion capacitance appears in series with oxide capacitance, so total capa |
| Inversion (VG sufficiently positive) | Electrons form an inversion channel | At low frequency, inversion charge can respond and capacitance rises toward |

### 3. Oxide and depletion capacitances

The oxide capacitance and depletion capacitance are in series during depletion. The equivalent capacitance per unit area is:

**Ctotal = (Cox × Cdep) / (Cox + Cdep)**

Since a series combination is smaller than either individual capacitance, Ctotal becomes less than Cox during depletion. When the depletion region becomes wider, Cdep decreases and the measured total capacitance decreases further.

In strong inversion, the measured result depends on measurement frequency and whether inversion charge can respond. Low-frequency C–V measurement generally returns toward Cox; high-frequency measurement may stay near the minimum depletion capacitance.

### 4. PMOS capacitance behavior

For a PMOS built in an n-well, the substrate/body type and voltage polarities are reversed compared with the NMOS example. Accumulation occurs when the gate bias attracts the majority carriers of the n-well surface; depletion occurs when they are pushed away; inversion occurs when a p-type surface channel is formed. The same oxide/depletion series-capacitance principle applies, with voltage polarity reversed.

**Interpretation:** The gate-voltage-dependent capacitance curve reveals whether the surface is in accumulation, depletion or inversion. Oxide thickness, device area, doping and measurement frequency affect the curve.

## Part (c) – CMOS Inverter Voltage-Transfer Characteristic

### 1. CMOS inverter structure

A CMOS inverter contains one PMOS transistor at the top connected to VDD and one NMOS transistor at the bottom connected to ground. Their gates are joined to form the input Vin, and their drains are joined to form the output Vout.

The PMOS pulls the output HIGH when the input is LOW. The NMOS pulls the output LOW when the input is HIGH. The output is the logical opposite of the input.

### 2. Voltage-transfer characteristic (VTC)

The VTC is a graph of output voltage Vout against input voltage Vin. It has a high-output region, a steep transition region and a low-output region.

| Input range / condition | NMOS state | PMOS state | Output |
|---|---|---|---|
| Vin near 0 V | Cut-off | ON (often linear near the rail) | Vout ≈ VDD |
| Vin increasing, below switching region | Begins to conduct / linear | ON | Vout remains high but starts falling |
| Vin near switching threshold VM | Conducting; often saturation | Conducting; often saturation | Vout changes sharply |
| Vin above switching region | ON | Becomes weaker, then cut-off | Vout approaches 0 V |
| Vin near VDD | ON (often linear near ground) | Cut-off | Vout ≈ 0 V |

### 3. Important transition points

• VOH: minimum output voltage guaranteed as logic HIGH; for an ideal unloaded CMOS inverter it is close to VDD.

• VOL: maximum output voltage guaranteed as logic LOW; for an ideal unloaded inverter it is close to 0 V.

• VM: switching threshold where Vin = Vout. It is near VDD/2 for a suitably balanced inverter, but the exact value depends on transistor strengths and thresholds.

• VIL: largest input voltage still recognised as LOW.

• VIH: smallest input voltage recognised as HIGH.

• Noise margins: NML = VIL – VOL and NMH = VOH – VIH.

### 4. Transistor operation during transition

At a LOW input, the PMOS is ON and the NMOS is OFF, so there is a path from VDD to the output and Vout is HIGH. As Vin rises above the NMOS threshold, the NMOS begins conducting. In the transition region, both devices can conduct simultaneously and the output falls rapidly. At a HIGH input, the NMOS is ON and the PMOS is OFF, so the output is pulled to ground.

During the short switching interval, both transistors may conduct, creating a temporary direct current path from VDD to ground. This is called short-circuit or crowbar current. It is reduced by choosing suitable transistor sizes and input transition times.

**Switching interpretation:** The steep VTC transition provides good noise immunity because a small input change near VM causes a large output change. The exact VTC depends on VDD, threshold voltages, W/L ratios, mobility and load capacitance.

## Part (d) – N-well CMOS Structure and Operation

### 1. N-well fabrication structure

In a common n-well CMOS process, a lightly doped p-type silicon substrate is used. An n-well is formed inside it. The PMOS transistor is fabricated inside the n-well, while the NMOS transistor is fabricated directly in the p-type substrate.

• The NMOS uses n+ source and drain regions in the p-type substrate.

• The PMOS uses p+ source and drain regions inside the n-well.

• A thin gate oxide separates each gate electrode from the silicon channel region.

• The NMOS body is normally connected to ground (VSS).

• The PMOS n-well body is normally connected to the positive supply (VDD).

• Isolation structures separate neighbouring devices and help prevent unwanted electrical paths.

### 2. Simplified CMOS inverter circuit

| Connection | Description |
|---|---|
| PMOS source and body | Connected to VDD |
| NMOS source and body | Connected to GND / VSS |
| Both gates | Connected together as Vin |
| Both drains | Connected together as Vout |

**Conceptual structure (not to scale):**

| VDD |
|---|
| ■ |
| PMOS (inside n-well) |
| ■ |
| ● Vout ■■■ Output |
| ■ |
| NMOS (in p-substrate) |
| ■ |
| GND |

The input Vin connects to both transistor gates. The output Vout is the common connection between the two drains.

### 3. Complementary switching

| Input Vin | PMOS | NMOS | Output Vout |
|---|---|---|---|
| Logic LOW (0) | ON | OFF | Logic HIGH (1) |
| Logic HIGH (1) | OFF | ON | Logic LOW (0) |

For logic LOW, the gate voltage is low: the PMOS gate-to-source voltage is sufficiently negative to turn it ON, while the NMOS gate-to-source voltage is below threshold and it is OFF. The output is pulled up to VDD.

For logic HIGH, the NMOS gate-to-source voltage is sufficiently positive to turn it ON, while the PMOS gate-to-source voltage is too small in magnitude to remain ON. The output is pulled down to ground.

**Why it is called complementary:** the two devices respond in opposite ways to the same input. In steady logic states, ideally one transistor is OFF, so there is almost no direct DC path from VDD to ground.

## Part (e) – Low-Power Performance Evaluation

### 1. Current-voltage behavior and switching

The I–V characteristics show how much current a transistor can supply at a particular gate and drain voltage. A suitable NMOS–PMOS pair must provide enough drive current to charge and discharge the load capacitance within the required time. Stronger drive can improve speed, but larger devices also add capacitance and area.

### 2. Capacitance and dynamic power

The gate and other internal capacitances must be charged and discharged during switching. Dynamic power is approximately:

**$P_{dynamic} = \alpha C_{load} V_{DD}^2 f$**

where $\alpha$ is the switching activity factor, Cload is the effective switched capacitance, VDD is the supply voltage and f is the switching frequency. Lower capacitance, lower supply voltage and lower unnecessary switching reduce dynamic power.

### 3. Static and short-circuit power

In a stable CMOS logic state, ideally one transistor is OFF and the direct current from VDD to ground is nearly zero. Real devices still have leakage current, so static power is approximately Pstatic = Ileak × VDD. During transitions both transistors can briefly conduct, causing short-circuit power.

### 4. Noise immunity and voltage transfer

A steep inverter VTC creates a clear distinction between logic LOW and logic HIGH. Noise margins NMH and NML show how much unwanted voltage disturbance can be tolerated. A well-designed inverter has adequate margins across supply, process and temperature variations.

### 5. Overall suitability

| Characterization result | Effect on performance | Design interpretation |
|---|---|---|
| I–V curve and operating regions | Determines drive current and switching speed | Select W/L sizes to meet load and delay requirements. |
| Gate capacitance vs. voltage | Affects input loading and energy per transition | Avoid unnecessarily large gate area and capacitive loads. |
| Steep VTC and noise margins | Improves logic recognition and noise immunity | Maintain adequate HIGH and LOW margins. |
| Complementary NMOS/PMOS action | Very low ideal steady-state current | Suitable for low-power digital logic. |
| Leakage and transition current | Contribute to static and dynamic losses | Use suitable process, supply and switching frequency. |

**Evaluation:** The NMOS–PMOS pair is highly suitable for low-power digital switching because complementary operation ideally turns one device OFF in each stable state. However, actual power and speed depend on leakage, load capacitance, supply voltage, device dimensions, switching activity and transition time.

## Conclusion

NMOS and PMOS transistors operate in cut-off, linear and saturation regions according to their gate and drain voltages. Their gate capacitance changes as the surface moves through accumulation, depletion and inversion. In a CMOS inverter, the PMOS and NMOS switch in opposite ways, producing an inverted output with a steep voltage-transfer curve. An n-well process places PMOS devices in an n-well and NMOS devices in a p-type substrate. The complementary arrangement gives low steady-state power and good noise immunity, making CMOS suitable for low-power digital systems.

## References


• ECA02 Digital Circuits – CO5 Characteristics Interpretation Assignment supplied for this work.

• Standard long-channel MOSFET I–V equations and MOS capacitor C–V concepts from digital electronics / semiconductor device coursework.

• Device and process datasheets or SPICE model documentation should be used for actual threshold voltage, mobility, oxide thickness, capacitance, leakage and timing values.


**Submission note:** The supplied assignment does not include measured characterization plots or device parameters. This answer therefore explains the expected standard behavior. Add measured I–V, C–V and inverter VTC plots if your instructor expects experimental data.